THE FITTING IDEAL OF THE STRICT TRANSFORM ALONG THE BLOWUP IN THAT IDEAL

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Formation of Fitting ideals commutes with base change. Lemma \ref{lemma-fitting-ideal-basics}, cited at proof lines 6186-6187. Identifies the Fitting ideal of the base changed module.
- The blowup algebra makes the extended ideal principal. Algebra, Lemma \ref{algebra-lemma-affine-blowup}, cited at proof lines 6187-6188. Turns the extended Fitting ideal into a principal ideal.
- Local characterisation of the top Fitting ideal, and the description of a principal Fitting ideal. Lemma \ref{lemma-principal-fitting-ideal} and Lemma \ref{lemma-fitting-ideal-generate-locally}, cited at proof lines 6189-6195. Convert the local generation bound into the statement that the top Fitting ideal is the unit ideal.

Source reference. Original lemma 0CZM: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 16282–16282. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 07ZA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10140–10140. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07Z3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10133–10133. Navigation pointer to a statement the

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Source reference. Original lemma 080Z: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10199–10199. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 07ZC: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10142–10142. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let R be a ring. Let M be a finite R-module. Let $k \geq 0$ and $I = Fit_k(M)$. For every $a \in I$ with $R' = R[\frac{I}{a}]$ the strict transform*

$$M' = (M \otimes_R R')/a\text{-power torsion}$$

has $Fit_k(M') = R'$.

Proof. First observe that $\mathrm{Fit}_k(M \otimes_R R') = IR' = aR'$. The first equality by Lemma 07ZA part (3) and the second equality by Algebra, Lemma 07Z3. From Lemma 080Z and exactness of localization we see that $M'_{\mathfrak{p}'}$ can be generated by $\leq k$ elements for every prime $\mathfrak{p}'$ of R'. Then $\mathrm{Fit}_k(M') = R'$ for example by Lemma 07ZC. □